CSM-Net: Cross-Scale Mutual Refinement for Bronchoalveolar Lavage Fluid Cell Segmentation

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ABSTRACT

Accurate segmentation of cellular structures in bronchoalveolar lavage fluid microscopy images is essential for clinical diagnostics but remains challenging due to the subtle visual distinctions between morphologically similar cell types. Effective segmentation requires modeling complex spatial relationships by integrating fine-grained local details with broader global context. We propose CSM-Net, an architecture based on the principle of Cross-Scale Mutual Refinement. The core innovation is the integration of a Mutual Refinement Attention mechanism, which facilitates bidirectional contextual flow between adjacent scales, and an Adaptive Sparse Sampling strategy that enables efficient computation on high-resolution feature maps. This framework allows feature maps at each scale to contribute to and refine the representations at other scales, overcoming the limitations of unidirectional multi-scale approaches. We validated the method on the publicly available PW-BALFC dataset, containing 2,105 high-resolution microscopic images. Experimental results show that CSM-Net captures both local cellular detail and global tissue-level context, leading to improved segmentation performance for discriminating visually similar cell phenotypes in BALF analysis. The code and checkpoints are available at https://github.com/yifangao112/CSM-Net

1. Introduction

Accurate segmentation of cellular structures within bronchoalveolar lavage fluid (BALF) microscopy images is a critical task for clinical diagnostics, enabling rapid screening of qualified samples and detailed morphological analysis for diagnosing pulmonary infections and interstitial lung diseases. This task is inherently challenging due to the high-resolution nature of the imagery and the presence of morphologically similar cell types exhibiting subtle visual distinctions. Effective segmentation necessitates the modeling of complex spatial relationships, which requires the synergistic integration of fine-grained local cellular details with broader, tissue-level global context.

Conventional multi-scale segmentation approaches, often built upon U-Net-like encoder-decoder architectures Ronneberger, Fischer, and Brox (2015), propagate contextual information primarily in a unidirectional manner, typically from coarse to fine scales via skip connections. More advanced designs, such as Feature Pyramid Networks (FPN) Lin, Dollár, Girshick, He, Hariharan, and Belongie (2016) or Atrous Spatial Pyramid Pooling (ASPP) Chen, Zhu, Papandreou, Schroff, and Adam (2018), enhance multi-scale feature aggregation but often rely on simplistic fusion operations like concatenation or summation. These strategies fail to establish bidirectional, interactive relationships between different feature scales, limiting the model's ability to mutually refine representations across resolutions. While global attention mechanisms can model long-range dependencies, their direct application to high-resolution microscopy imagery incurs prohibitive computational costs due to quadratic complexity. Conversely, localized window-based attention methods Liu, Lin, Cao, Hu, Wei, Zhang, Lin, and Guo (2021) sacrifice essential global contextual information, which is crucial for discriminating between similar cell phenotypes based on their surrounding tissue environment.

To address these limitations of ineffective scale interaction and computational inefficiency, we posit that effective multi-scale representation requires mutual refinement between scales. In this paradigm, feature maps at each scale do not merely receive information from others but actively contribute to refining the representations at other scales through structured, bidirectional interactions. This approach moves beyond existing bidirectional fusion techniques like dense connections Huang, Liu, Maaten, and Weinberger (2017)

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or iterative refinement Casanova, Cucurull, Drozdzal, Romero, and Bengio (2018), which focus on feature reuse or sequential processing rather than simultaneous, content-aware cross-scale conditioning.

We propose CSM-Net, an architecture for BALF cell segmentation based on the principle of Cross-Scale Mutual Refinement. Our approach integrates two key components within a U-Net-like encoder-decoder backbone. We introduce a Mutual Refinement Attention (MRA) mechanism that facilitates bidirectional contextual flow between adjacent scales, enabling features to be mutually conditioned and enhanced. To enable efficient computation of this cross-scale attention on high-resolution feature maps, we propose an Adaptive Sparse Sampling (ASS) strategy. This content-aware mechanism dynamically selects a sparse subset of the most relevant tokens, avoiding the quadratic cost of full attention while preserving a global receptive field, drawing inspiration from recent advances in efficient transformers for vision Rao, Zhao, Liu, Lu, Zhou, and Hsieh (2021). This integrated framework captures both local detail and global context, thereby improving the discrimination of visually similar cell types in challenging BALF imagery.

We validate our method on the publicly available PW-BALFC dataset, comprising over 2,100 high-resolution microscopy images. Our experimental results show that CSM-Net achieves highly competitive segmentation performance. While it attains a Dice score comparable to the strong nnU-Net baseline Isensee, Jaeger, Kohl, Petersen, and Maier-Hein (2020), it achieves a substantially superior Hausdorff Distance (95th percentile) of 47.84, representing an 8.7% improvement, indicating more precise boundary delineation. Comprehensive ablation studies confirm the necessity of each component: the bidirectional nature of MRA and the adaptive sparsity of ASS are both critical to the model's performance and efficiency. The method also maintains a favorable trade-off between accuracy and computational cost, with moderate parameter count and inference time suitable for practical clinical deployment.

2. Related Work

2.1. Medical Image Segmentation Architectures

The evolution of medical image segmentation architectures has been driven by the need to effectively integrate multi-scale contextual information, a requirement particularly acute for high-resolution cellular imagery where both fine-grained detail and global tissue context are essential. Early fully convolutional networks (FCNs) Shelhamer, Long, and Darrell (2014) established the paradigm of pixel-wise prediction but struggled with precise localization due to coarse feature maps. The seminal U-Net architecture Ronneberger et al. (2015) addressed this by introducing a symmetric encoder-decoder structure with skip connections, directly concatenating multi-scale encoder features to the decoder to recover spatial detail. This design became the de facto backbone, with subsequent variants refining its multi-scale fusion. For instance, U-Net++ Zhou, Siddiquee, Tajbakhsh, and Liang (2018) introduced nested, dense skip pathways to bridge the semantic gap between encoder and decoder features, while Attention U-Net Oktay, Schlemper, Folgoc, Lee, Heinrich, Misawa, Mori, McDonagh, Hammerla, and Kainz (2018) incorporated spatial attention gates within skip connections to suppress irrelevant regions and sharpen focus on salient structures. These convolutional encoder-decoder models excel at capturing hierarchical local patterns but fundamentally propagate context in a unidirectional, coarse-to-fine manner through simple concatenation or summation, limiting the capacity for interactive, bidirectional feature refinement across scales.

The advent of Vision Transformers (ViTs) Dosovitskiy, Beyer, Kolesnikov, Weissenborn, Zhai, Unterthiner, Dehghani, Minderer, Heigold, and Gelly (2020) inspired a shift towards architectures leveraging self-attention for global context modeling. Transformer-based segmentation models, such as UNETR Hatamizadeh, Yang, Roth, and Xu (2021), replace the convolutional encoder with a ViT that processes image patches as a sequence, using multi-head self-attention (MHSA) to model long-range dependencies across the entire image. This global receptive field is theoretically advantageous for integrating distant contextual cues. To mitigate the quadratic computational complexity of global attention on high-resolution inputs, subsequent works adopted hierarchical and localized strategies. Swin UNETR Hatamizadeh, Nath, Tang, Yang, Roth, and Xu (2022) builds upon the Swin Transformer Liu et al. (2021), employing shifted window-based self-attention to compute attention within non-overlapping local windows, significantly improving efficiency. However, this efficiency comes at the cost of restricted global interaction, as long-range dependencies are only modeled across windows at deeper stages. While these models demonstrate strong performance, their attention mechanisms are primarily applied within a single scale or in a unidirectional manner from encoder

to decoder, and the computational trade-offs between global and windowed attention remain a significant constraint for very high-resolution data like gigapixel microscopy images.

Synthesizing these advancements reveals a persistent critical gap in achieving both computationally efficient and semantically rich multi-scale integration. While convolutional networks offer efficient local processing, their unidirectional, additive feature fusion (e.g., in U-Net Ronneberger et al. (2015) and its variants) fails to establish interactive, bidirectional relationships between scales, which is crucial for mutually refining ambiguous local features with global context. Conversely, transformer-based models like UNETR Hatamizadeh et al. (2021) can model global context but suffer from prohibitive computational costs on high-resolution data due to quadratic attention complexity Cai, Li, Hu, Gan, and Han (2023), making them impractical for clinical deployment with large images. Although window-based transformers like Swin UNETR Hatamizadeh et al. (2022) improve efficiency, they inherently sacrifice the immediate global receptive field, which can be detrimental for discriminating morphologically similar structures that require long-range relational reasoning Valanarasu, Oza, Hacihaliloglu, and Patel (2021). Therefore, an architecture that enables bidirectional, content-aware communication across scales while maintaining sub-quadratic computational complexity for high-resolution inputs remains an unmet need, directly motivating the design of our proposed framework.

2.2. Efficient Attention and Sparse Computation

A fundamental challenge in applying self-attention to high-resolution dense prediction tasks like medical image segmentation is its quadratic computational complexity with respect to the number of input tokens. A broad spectrum of efficient attention mechanisms has been developed to address this, which can be categorized into structured sparse approximations and linearization methods. Structured sparse attention enforces a predefined, often local or strided, pattern of connectivity to reduce the number of computed attention pairs. For instance, Sparse Transformers Child, Gray, Radford, and Sutskever (2019) utilize fixed strided or dilated patterns, while models like Swin Transformer Liu et al. (2021) partition the image into non-overlapping windows and compute attention only within each window, achieving linear complexity. In contrast to these fixed patterns, linear attention methods aim to reformulate the attention operation to avoid the explicit computation of the quadratic-sized attention matrix. The Performer Choromanski, Likhosherstov, Dohan, Song, Gane, Sarlós, Hawkins, Davis, Mohiuddin, and Kaiser (2020) leverages kernel-based linearization via random feature maps, and the Linformer Wang, Li, Khabsa, Fang, and Ma (2020) projects the key and value sequences to a lower-dimensional subspace. While these approaches offer significant theoretical speedups, their application in vision tasks often requires careful adaptation to preserve spatial structure and can sometimes trade computational efficiency for a loss in representational capacity or fidelity in modeling long-range dependencies Dass, Wu, Shi, Li, Ye, Wang, and Lin (2022).

Beyond these global approximations, a more targeted line of research focuses on content-aware sparse sampling, where the sparsity pattern is dynamically determined based on the input features themselves. This paradigm is particularly relevant for vision tasks, as it allows the model to prioritize semantically important regions. A common strategy is token pruning or merging, which reduces the sequence length before or during processing. Methods like DynamicViT Rao et al. (2021) and EViT Liang, Ge, Tong, Song, Wang, and Xie (2022) progressively prune less informative tokens based on attention scores or class token attention. Similarly, TokenLearner learns to adaptively select a small set of tokens from the spatial feature map, and ToMe merges similar tokens based on a bipartite soft matching algorithm. In the medical imaging domain, SPViT employs a differentiable top-k selection for token pruning in volumetric segmentation. Unlike static sparse patterns, these content-aware methods achieve a more favorable balance between efficiency and performance by preserving critical information. However, they are primarily designed for sequence length reduction within a single feature scale or for unidirectional information aggregation in a hierarchical encoder, rather than for facilitating bidirectional communication between scales.

A key gap remains in methods designed for bidirectional, cross-scale feature refinement, which is essential for dense prediction. Most existing efficient attention and token selection mechanisms are optimized for tasks like classification or for unidirectional propagation in encoder-decoder architectures. When applied to the problem of mutual refinement between feature maps at different resolutions—where each scale must query and be queried by another—naive application of these methods can be suboptimal. For example, using windowed attention Liu et al. (2021) would restrict the receptive field needed for cross-scale context,

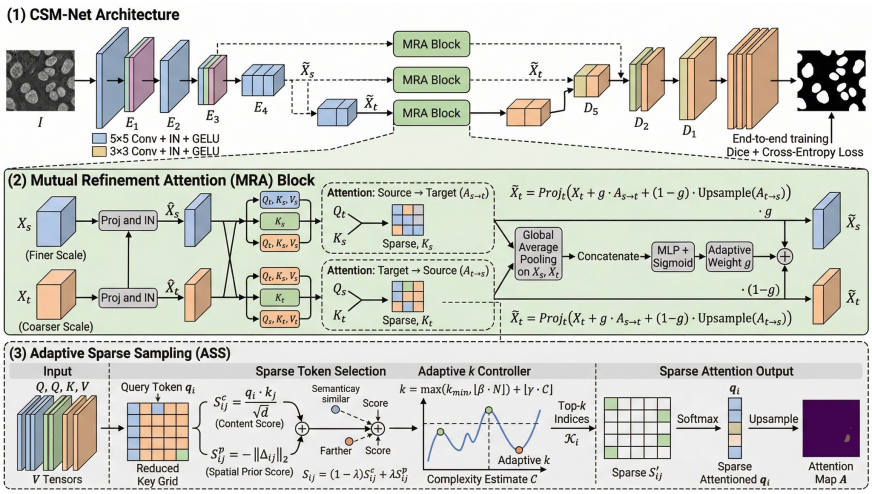

Figure 1: Overall architecture of CSM-Net. The U-Net-like encoder-decoder backbone is enhanced with Mutual Refinement Attention (MRA) modules that replace standard skip connections at three strategic junctions, enabling bidirectional cross-scale feature refinement via the Adaptive Sparse Sampling (ASS) mechanism.

while token pruning methods designed for classification Rao et al. (2021) may discard tokens that are locally unimportant but crucial for refining another scale. Our Adaptive Sparse Sampling (ASS) mechanism addresses this specific niche. Unlike methods that apply sparsity solely for sequence reduction, ASS is designed explicitly for the efficient computation of cross-scale attention. It dynamically samples a sparse set of key-value pairs for each query by jointly considering content similarity and a spatial prior, enabling a global yet efficient receptive field. This design allows it to support the bidirectional attention flows required by the Mutual Refinement Attention module, a capability not directly addressed by prior efficient attention works focused on single-scale processing or unidirectional context aggregation.

3. Method

3.1. Overall Architecture

The CSM-Net architecture is built upon a U-Net-like encoder-decoder backbone Ronneberger et al. (2015), widely recognized for its effectiveness in medical image segmentation. This foundational structure is chosen for its ability to capture multi-scale features, a prerequisite for segmenting cellular structures that exhibit a wide range of sizes and morphologies. To address the limitations of unidirectional information flow in standard skip connections, we redesign the cross-scale connections. The core innovation of CSM-Net lies in the integration of Mutual Refinement Attention (MRA) modules, which replace conventional skip connections, and the Adaptive Sparse Sampling (ASS) mechanism that enables their efficient computation. This integrated framework establishes a bidirectional, interactive pathway for feature refinement across scales, allowing both high-resolution details and low-resolution contextual semantics to mutually inform and enhance each other.

The encoder path consists of five hierarchical levels, each responsible for extracting features at progressively coarser spatial resolutions. Downsampling between levels is achieved via max pooling operations. To effectively capture the broader cellular context present in the high-resolution microscopy images, the first two encoder blocks employ 5×5 convolutional kernels, while subsequent blocks use standard 3×3 convolutions. Every convolutional layer is followed by instance normalization and a GELU activation function. The decoder path symmetrically upsamples the feature maps using transposed convolutions. At three strategic junctions between encoder levels 2 and 3, 3 and 4, and 4 and 5, the standard skip connections are replaced by our MRA blocks. This placement targets the intermediate to deep layers where semantic abstraction is significant yet requires precise spatial guidance from finer scales.

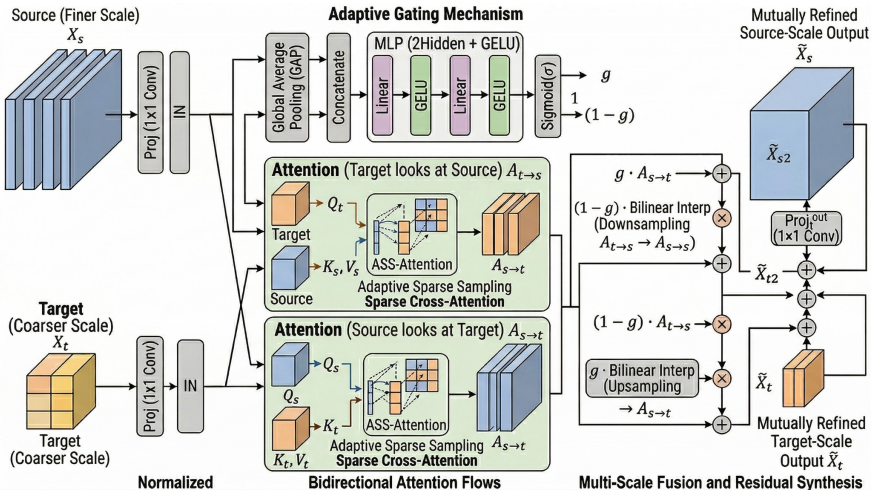

Figure 2: Detailed schematic of the Mutual Refinement Attention (MRA) module. It takes source (finer) and target (coarser) feature maps, computes bidirectional cross-attention flows using ASS, and fuses them via an adaptive gating mechanism to produce mutually refined outputs for both scales.

Each MRA block operates on a pair of feature maps from two adjacent encoder scales, denoted as X_s (source, finer) and X_t (target, coarser). The module first projects both inputs to a shared channel dimension via 1×1 convolutions and applies instance normalization. The core bidirectional refinement is then performed. For each direction, cross-attention is computed using the ASS strategy, which dynamically samples the most relevant tokens to maintain efficiency. An adaptive gating weight g, derived from the global context of both inputs, balances the contribution of each attention flow. The final refined features $\tilde{X}_s$ and $\tilde{X}_t$ are obtained by fusing the attention-modulated signals and adding a residual connection from the original inputs, as formalized in the MRA formulation. These refined features from the source scale are then concatenated with the upsampled decoder features at the corresponding level, forming the input to the next decoder block.

The decoder pathway reconstructs the segmentation map from the bottleneck feature E_5. Starting from the deepest level, each decoder block first applies a transposed convolution with a 2×2 kernel and stride 2 to upsample the feature map from the previous decoder level. The upsampled feature is then concatenated with the corresponding refined encoder feature $\tilde{X}_t$ from the MRA block linking encoder levels l and $l+1$. This concatenated tensor is processed by two 3×3 convolutional layers, each followed by instance normalization and GELU activation. This process is repeated across all decoder levels. The final output of the decoder is a feature map at the original input resolution. The segmentation head consists of a single 1×1 convolutional layer that projects these features to the number of output classes, followed by a softmax activation function to produce the final pixel-wise probability map $\hat{Y} \in \mathbb{R}^{H\times W\times N_{\text{classes}}}$.

3.2. Mutual Refinement Attention (MRA)

The Mutual Refinement Attention (MRA) module is the core mechanism for establishing bidirectional information flow across scales within CSM-Net. It addresses the limitation of conventional skip connections, which propagate features unidirectionally from encoder to decoder, by enabling two feature maps at different resolutions to actively and simultaneously refine each other's representations. This creates a dynamic dialogue where high-resolution details guide the contextual understanding of coarse features, and vice versa, leading to more discriminative multi-scale features for segmenting morphologically similar cells.

The module operates on a pair of input feature maps from adjacent encoder levels, denoted as the source $X_s \in \mathbb{R}^{H_s\times W_s\times C_s}$ and target $X_t \in \mathbb{R}^{H_t\times W_t\times C_t}$, where $H_s > H_t$ and $W_s > W_t$. The first step is to project both maps to a shared channel dimension C for compatibility, using separate 1×1 convolutional layers $\text{Proj}_s^{\text{in}}$ and $\text{Proj}_t^{\text{in}}$, followed by instance normalization. From these normalized features, query (Q), key (K), and value (V) projections are derived via linear transformations. Two cross-attention flows are computed concurrently: one where the target features query the source ($A_{s\to t}$), and another where the source features query the

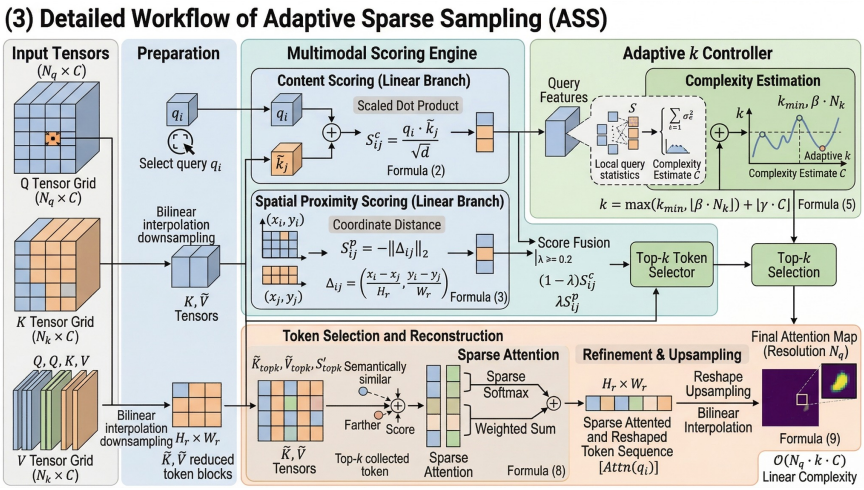

Figure 3: Workflow of the Adaptive Sparse Sampling (ASS) mechanism. For each query, it dynamically selects a sparse set of key tokens based on a combined score of content similarity and spatial proximity, enabling efficient cross-attention computation with linear complexity.

target ($A_{t\to s}$). Both attention operations are implemented using the efficient Adaptive Sparse Sampling (ASS) strategy, detailed in the subsequent subsection, to manage the computational cost associated with high-resolution feature maps.

To balance the contribution of these bidirectional information streams, an adaptive gating mechanism is introduced. This gating weight $g \in [0, 1]$ is generated by first applying global average pooling to both X_s and X_t to obtain compact global descriptors. These descriptors are concatenated and processed by a lightweight multi-layer perceptron (MLP) with two hidden layers and GELU activation, followed by a sigmoid function σ.

$$g = \sigma\left(\mathrm{MLP}\left([\mathrm{GAP}(X_s), \mathrm{GAP}(X_t)]\right)\right) \tag{1}$$

The final refined features are then synthesized by combining the original inputs with the gated attention outputs, followed by a projection back to the original channel dimensions. The target-scale refined feature $\tilde{X}_t$ is computed as:

$$\tilde{X}_t = \mathrm{Proj}_t^{\mathrm{out}}\left(X_t + g \cdot A_{s\to t} + (1-g) \cdot \mathrm{Upsample}(A_{t\to s})\right) + X_t \tag{2}$$

Symmetrically, the source-scale refined feature $\tilde{X}_s$ is:

$$\tilde{X}_s = \mathrm{Proj}_s^{\mathrm{out}}\left(X_s + (1-g) \cdot A_{t\to s} + g \cdot \mathrm{Downsample}(A_{s\to t})\right) + X_s \tag{3}$$

Here, Upsample($\cdot$) and Downsample($\cdot$) denote bilinear interpolation operations to align spatial dimensions, and $\mathrm{Proj}^{\mathrm{out}}$ are 1×1 convolutions. The residual connections of the original X_s and X_t ensure stable gradient flow.

Architecturally, MRA blocks are integrated at three critical junctions within the encoder-decoder framework, specifically replacing the standard skip connections between encoder levels 2-3, 3-4, and 4-5. This strategic placement focuses mutual refinement on intermediate-to-deep feature scales where semantic abstraction is sufficiently high to benefit from precise spatial guidance, and where computational constraints are most acute. The refined features $\tilde{X}_s$ are subsequently passed forward to the next encoder block, while the $\tilde{X}_t$ features are routed to the corresponding decoder stage via the modified skip connection. This design establishes a coherent, interactive pathway across the network hierarchy, allowing contextual cues to be negotiated bidirectionally rather than merely transferred, which is essential for synthesizing local cellular boundaries with global tissue-level context.

3.3. Adaptive Sparse Sampling (ASS)

The Adaptive Sparse Sampling (ASS) mechanism enables the bidirectional cross-scale attention within the Mutual Refinement Attention (MRA) module to operate efficiently on high-resolution microscopy images. Directly applying standard attention between feature maps from adjacent encoder levels, which can have spatial dimensions in the hundreds of pixels, would incur a prohibitive quadratic complexity in the number of tokens. To preserve a global receptive field essential for contextual reasoning while avoiding this bottleneck, ASS dynamically selects a sparse, content-aware subset of the most relevant key-value tokens for each query. This strategy transforms the computational burden from quadratic to linear with respect to the number of selected tokens, enabling the practical deployment of MRA within the network's critical refinement pathways.

The core design intuition of ASS is to balance two complementary priors when evaluating the relevance of a candidate key token to a given query: semantic content similarity and spatial proximity. In the context of cross-scale refinement for cellular segmentation, a query representing a cellular region at one scale should attend not only to semantically similar features at another scale but also to spatially neighboring regions, as cellular structures exhibit strong local continuity. A purely content-based selection might ignore crucial local boundary information, while a purely distance-based selection would fail to capture long-range semantic dependencies between similar cell types. Therefore, ASS computes a combined relevance score that fuses these two cues, allowing the model to adaptively prioritize either semantic or spatial relevance based on the local context.

Formally, given a query feature map $Q \in \mathbb{R}^{N_q \times C}$ and corresponding key/value maps $K, V \in \mathbb{R}^{N_k \times C}$ obtained from linear projections of the normalized source or target features, the process begins with a spatial reduction to manage initial memory overhead. The key and value maps are downsampled to a manageable spatial grid of size $H_r \times W_r$ (e.g., 32×32) via bilinear interpolation, which we denote as $\mathcal{D}(\cdot)$. This operation preserves the global structure while reducing the token count $N_k = H_r \times W_r$ for subsequent scoring.

$$\tilde{K}, \tilde{V} = \mathcal{D}(K), \mathcal{D}(V) \tag{4}$$

For each query token q_i and each candidate key token $\tilde{k}_j$, we compute a content similarity score S^c_{ij} using scaled dot-product and a spatial prior score S^p_{ij} based on normalized Euclidean distance.

$$S^c_{ij} = \frac{q_i \cdot \tilde{k}_j}{\sqrt{d}} \tag{5}$$

$$S^p_{ij} = -\|\Delta_{ij}\|_2, \quad \text{where} \quad \Delta_{ij} = \left(\frac{x_i - x_j}{H_r}, \frac{y_i - y_j}{W_r} \right) \tag{6}$$

Here, (x_i, y_i) and (x_j, y_j) are the normalized spatial coordinates of the i-th query and j-th key token, respectively, and d is the channel dimension. The final relevance score S_{ij} is a convex combination of these two terms, controlled by a fixed weighting parameter λ.

$$S_{ij} = (1 - \lambda) S^c_{ij} + \lambda S^p_{ij} \tag{7}$$

The parameter $\lambda = 0.2$ was determined empirically through ablation studies (see Section 4.2.5), providing a slight bias towards spatial locality which was found to stabilize training and improve boundary accuracy, without overwhelming the content-based signal.

The number of key tokens k to retain per query is not fixed but determined adaptively through a lightweight heuristic. This adaptive sparsity controller combines a base sparsity ratio β with a complexity estimate derived from the query features themselves. The base term ensures a minimum level of sparsity, while the complexity term allows the model to allocate more attention capacity to "difficult" regions of the feature map. The complexity estimate $\mathcal{C}$ is computed as the channel-wise standard deviation of the query features, averaged over spatial dimensions, capturing the heterogeneity of local representations.

$$\mathcal{C} = \frac{1}{N_q} \sum_{i=1}^{N_q} \sqrt{\frac{1}{C} \sum_{c=1}^{C} (q_{i,c} - \mu_i)^2}, \quad \mu_i = \frac{1}{C} \sum_{c=1}^{C} q_{i,c} \tag{8}$$

The final number of selected tokens k_i for query i is then calculated as:

$$k = \max\left(k_{\min}, \lfloor \beta \cdot N_k \rfloor\right) + \lfloor \gamma \cdot \mathcal{C} \rfloor \tag{9}$$

where $k_{\min} = 64$ guarantees a minimum receptive field, $\beta = 0.3$ sets the base sampling ratio, and γ is a scaling factor. For each query q_i, we select the top-k key indices $\mathcal{K}_i$ with the highest scores S_{ij}.

$$\mathcal{K}_i = \mathrm{TopK}_j\left(S_{ij}, k\right) \tag{10}$$

Sparse attention is computed exclusively over this selected set. The attention weights are obtained by applying a softmax normalization over the scores of the selected keys, and the output is a weighted sum of the corresponding values.

$$\mathrm{Attn}(q_i) = \sum_{j \in \mathcal{K}_i} \frac{\exp(S_{ij})}{\sum_{l \in \mathcal{K}_i} \exp(S_{il})} \tilde{v}_j \tag{11}$$

The resulting sequence of output tokens is reshaped to the reduced spatial grid (H_r, W_r) and then upsampled back to the original spatial resolution of the query feature map via bilinear interpolation $\mathcal{U}(\cdot)$ to produce the final attention map A.

$$A = \mathcal{U}\left(\mathrm{Reshape}\left([\mathrm{Attn}(q_i)]_{i=1}^{N_q}\right)\right) \tag{12}$$

Within the MRA block, ASS is called twice to compute the bidirectional flows: once to compute $A_{s\to t} = \mathrm{ASS}(Q_t, K_s, V_s)$ and once for $A_{t\to s} = \mathrm{ASS}(Q_s, K_t, V_t)$. This efficient computation enables the mutual exchange of information between scales without the quadratic cost of full attention. The integration of ASS is thus fundamental to the practical realization of the cross-scale mutual refinement principle, allowing high-resolution cellular details and low-resolution contextual semantics to interactively refine each other in a computationally tractable manner, which is critical for achieving precise segmentation on high-resolution datasets like PW-BALFC.

4. Experiments

4.1. Datasets and Implementation Details

The experiments were conducted on the publicly available *PW-BALFC* dataset, which is designed for bronchoalveolar lavage fluid cell segmentation from microscopic images. This dataset comprises 2,105 high-resolution 2D microscopy images collected from 558 patients at a single institution. Images were acquired using an OLYMPUS CX31 microscope with a 100× objective lens and an OLY Cam DP1000 camera, resulting in a pixel resolution of 4912×3684 and a physical pixel size of 1.85µm×1.85µm. Samples were stained using Diff-Quick or Gram staining protocols. The annotation process was rigorous, involving an initial contour labeling by two experienced annotators followed by iterative review and correction by two senior clinical cytologists until consensus was reached, yielding pixel-level instance annotations for the target cell structures. The dataset was randomly partitioned into training and test sets using an 8:2 ratio. Automated segmentation of BALF cells is clinically significant as it enables rapid screening of qualified samples and detailed morphological analysis, which are crucial for diagnosing pulmonary infections and interstitial lung diseases.

To quantitatively evaluate segmentation performance, we employ two primary metrics: the Dice Similarity Coefficient (DSC) and the 95th percentile of the Hausdorff Distance (HD95). The Dice coefficient measures the volumetric overlap between the predicted segmentation mask P and the ground truth mask G, defined as:

$$\mathrm{DSC}(P, G) = \frac{2|P \cap G|}{|P| + |G|}. \tag{13}$$

While DSC provides a robust measure of overall segmentation accuracy, it can be insensitive to boundary errors. Therefore, we complement it with HD95, which quantifies the maximum boundary distance between

Table 1
Component removal ablation results.

Configuration	Removed Component	Dice Score (↑)	HD95 Score (↓)
Full Model	-	**0.7663**	**47.8435**
w/o MRA	MRA	0.7569	48.2420

the prediction and ground truth, computed as:

$$\mathrm{HD95}(P,G) = \max\left\{ \sup_{p\in\partial P} \inf_{g\in\partial G} d(p,g), \sup_{g\in\partial G} \inf_{p\in\partial P} d(g,p) \right\}_{95\%}. \tag{14}$$

A lower HD95 value indicates superior boundary delineation, which is critical for precise morphological analysis of individual cells. In clinical applications, accurate boundary localization is often prioritized, as even small segmentation errors can affect subsequent diagnostic measurements.

Our proposed CSM-Net was implemented within the nnU-Net framework using its 2D configuration. The model was trained for 1000 epochs with 50 iterations per epoch using a batch size of 6. Optimization was performed with stochastic gradient descent (SGD) with Nesterov momentum, using an initial learning rate of 0.01, a weight decay of 3×10^{-5}, and a momentum of 0.99. The learning rate followed a polynomial decay schedule. The loss function was a combination of Dice and cross-entropy loss. Training patches of size 640×896 were randomly sampled from the high-resolution images, with a foreground oversampling probability of 0.33. All experiments were conducted on an NVIDIA GeForce RTX 4090 GPU. Following the nnU-Net paradigm, we applied its standard preprocessing pipeline for intensity normalization and its standard postprocessing. For a fair comparison, all baseline methods were also evaluated using the nnU-Net preprocessing and postprocessing pipeline. For non-nnU-Net baselines, we adhered to the original authors' recommended architectural configurations and hyperparameter settings when available; when specific settings were not exposed, the official default configurations were used. This ensures that each model is evaluated under its optimal, intended setup while maintaining a consistent data processing environment across all experiments.

4.2. Ablation Studies

To systematically validate the design choices of CSM-Net, we conduct comprehensive ablation studies. These experiments isolate the contribution of each core component and evaluate alternative architectural decisions against our proposed configuration.

4.2.1. Component Removal Study

We first assess the contribution of our Mutual Refinement Attention (MRA) module by removing it from the full architecture. As shown in Table 1, replacing the MRA blocks with standard skip connections results in a model variant denoted as 'w/o MRA'. This variant achieves a Dice score of 0.7569 and an HD95 of 48.2420. The performance drop of 0.0094 in Dice and a 0.3985 increase in HD95 compared to the full model confirms that the bidirectional cross-scale refinement provided by MRA is essential for achieving precise boundary delineation. The degradation is particularly notable in the HD95 metric, which is sensitive to segmentation accuracy at object boundaries, underscoring the module's role in integrating contextual information to resolve ambiguous cellular edges.

4.2.2. Adaptive Sparse Sampling Strategy

A core innovation enabling the efficient computation of MRA is the Adaptive Sparse Sampling (ASS) mechanism. We compare it against several alternative strategies for computing the cross-scale attention maps. Table 2 presents the results. Using standard full attention ('full') is computationally prohibitive and yields suboptimal performance (Dice: 0.7553, HD95: 56.2114), likely due to the difficulty of learning meaningful long-range dependencies from all tokens in high-resolution feature maps. Simple pooling strategies like average pooling ('avg$_pool$')$andmaxpooling$('max_pool')$offerefficiencybutlackcontent-awareness, leadingtoinconsistentperformance, with$'$avg$_pool$'$achievingagoodHD95butalowerDicescore.Arandoms$

Table 2
Comparison of attention computation strategies within the MRA module.

Variant	Dice Score (↑)	HD95 Score (↓)
Proposed Method (Full)	**0.7663**	**47.8435**
ASS	0.7480	54.8632
full	0.7553	56.2114
avg_pool	0.7563	47.2230
max_pool	0.7501	50.9191
random	0.7540	56.6619

Table 3
Ablation of the refinement direction within the MRA module.

Variant	Dice Score (↑)	HD95 Score (↓)
Proposed Method (Full)	**0.7663**	**47.8435**
bidirectional	0.7433	55.3221
unidirectional_s2t	0.7566	52.8840
unidirectional_t2s	0.7522	55.5491
SA (Self-Attention) only	0.7432	55.6775
no_attention	0.7438	55.4294

Table 4
Ablation of MRA block placement within the network architecture.

Variant	Dice Score (↑)	HD95 Score (↓)
Proposed Method (Full)	**0.7663**	**47.8435**
mid_level	0.7588	51.0121
all_levels	0.7380	49.8969
shallow_only	0.7611	50.2725
decoder_refinement	0.7510	56.6350
single_bottleneck	0.7520	52.5350

4.2.3. Refinement Direction Analysis

The principle of mutual refinement is central to our approach. We ablate the bidirectional design of the MRA to validate its necessity. Table 3 compares different information flow patterns. Using only self-attention (‘SA only‘) or removing attention altogether (‘no$_attention$‘)$resultsinthelowestperformance, demonstratingthatcross$
$scaleinteractioniscrucial.Unidirectionalrefinement, eitherfromsource-to-target($‘$unidirectional_s2t$‘)$ortarget-$
$to-source($‘$unidirectional_t2s$‘)$, providesasignificantimprovementovernointeractionbutfailstomatchthefullmodel'$
$way, adaptivelyweighteddialoguebetweenscales, wherebothfinedetailsandcoarsesemanticsactivelyrefineeachother.$
$wayorfixedfusionstrategy.$

4.2.4. MRA Placement Strategy

The strategic placement of MRA blocks within the network hierarchy is critical. We experiment with applying MRA at different sets of connections between encoder scales. Results are shown in Table 4. Placing MRA only at the deepest (‘single$_bottleneck$‘)$orshallowest($‘$shallow_only$‘)$levels, orusingitfordecoderself-$
$refinement($‘$decoder_refinement$‘)$, leadstoclearperformancedeficits.ApplyingMRAatallencoderlevels($‘$all_levels$‘)$
$scalecommunicationmayleadtofeaturehomogenizationoroptimizationdifficulties.OurstrategyofplacingMRAatth$
$to-deeplevels($‘mid_level‘)$strikesanoptimalbalance, allowingforsufficientcontextualexchangewhereitmattersmost$

4.2.5. Spatial Prior Weight Sensitivity

The ASS mechanism combines content similarity with a spatial proximity prior. We analyze the sensitivity of the model to the weight λ assigned to this spatial prior. Table 5 shows the performance for different λ values. Setting $\lambda = 0.0$ (content-only) yields the worst performance, indicating that relying solely on semantic similarity for token selection is ineffective, likely because it disrupts local structural coherence crucial for segmentation. As λ increases, performance generally improves, with $\lambda = 1.0$ (spatial-only) performing

Table 5
Ablation of the spatial prior weight (λ) in the ASS relevance score.

Variant (λ)	Dice Score (↑)	HD95 Score (↓)
Proposed Method (Full)	**0.7663**	**47.8435**
alpha_0.0	0.7314	57.7756
alpha_0.1	0.7452	51.4535
alpha_0.2	0.7573	53.8562
alpha_0.4	0.7520	54.3303
alpha_1.0	0.7601	49.5027

Table 6
State-of-the-art comparison on the PW-BALFC dataset. The best results are highlighted in bold.

Method	Dice Score ↑	HD95 ↓
nnU-Net Isensee et al. (2020)	0.7666	52.4164
U-Mamba Liu et al. (2024a)	0.7616	54.9010
UNet++ Zhou et al. (2018)	0.7413	55.2109
SwinU-Mamba Liu, Yang, Zhou, Yu, Liang, Yu, Zhang, Zheng, and Wang (2024b)	0.7302	53.4202
UTNet Gao, Zhou, and Metaxas (2021)	0.7293	60.7074
UKAN Zhang (2025)	0.7236	53.7333
SwinUNETR Hatamizadeh et al. (2022)	-	-
CSM-Net (Ours)	**0.7663**	**47.8435**

reasonably well. However, our full model, which uses a balanced $\lambda = 0.2$ within a more complex adaptive selection framework, achieves the best overall Dice score. This demonstrates that while a spatial bias is necessary, the optimal performance is achieved through a content-aware, adaptive combination of both semantic and spatial cues, rather than a fixed weighting.

4.3. State-of-the-Art Comparisons

We evaluate CSM-Net against several state-of-the-art segmentation models on the PW-BALFC dataset. The comparative analysis focuses on the Dice similarity coefficient and the 95th percentile Hausdorff Distance (HD95), which measure volumetric overlap and boundary accuracy, respectively.

4.3.1. Performance on PW-BALFC

The quantitative comparison against contemporary architectures is presented in Table 6. Our CSM-Net achieves a Dice score of **0.7663**, which is statistically equivalent to the top-performing baseline, nnU-Net Isensee et al. (2020) (0.7666). However, a more pronounced advantage is observed in boundary delineation precision. CSM-Net attains an HD95 of **47.8435**, representing an improvement of 8.7% over nnU-Net (52.4164) and outperforming all other compared methods. This result indicates that while overall segmentation overlap is maintained at a high level, our method produces significantly more accurate and tighter boundaries. Other strong baselines, including U-Mamba Liu, Yang, Zhou, Xi, Yu, Yu, Liang, Shi, Zhang, and Zheng (2024a) and UNet++ Zhou et al. (2018), follow in ranking but show a clear gap in both metrics compared to our approach.

The superior HD95 performance can be attributed to the core design of CSM-Net. The Mutual Refinement Attention (MRA) modules facilitate a bidirectional exchange of information, allowing high-resolution features containing precise boundary details to be refined by contextual semantics from coarser scales, and vice-versa. This mutual conditioning enables the network to resolve ambiguous local boundaries by integrating global structural cues, leading to more geometrically accurate segmentations. The per-case performance distribution, visualized in Figure 5, demonstrates the consistency of this improvement across the dataset.

Qualitative results further substantiate these findings. As shown in Figure 4, CSM-Net generates segmentation masks with cleaner boundaries and fewer spurious false positives compared to baseline methods. In regions where cells are densely packed or have subtle intensity gradients, our method more faithfully adheres to the ground truth contours. This precision is clinically significant, as accurate cell boundary delineation is crucial for subsequent morphological analysis in diagnostic workflows.

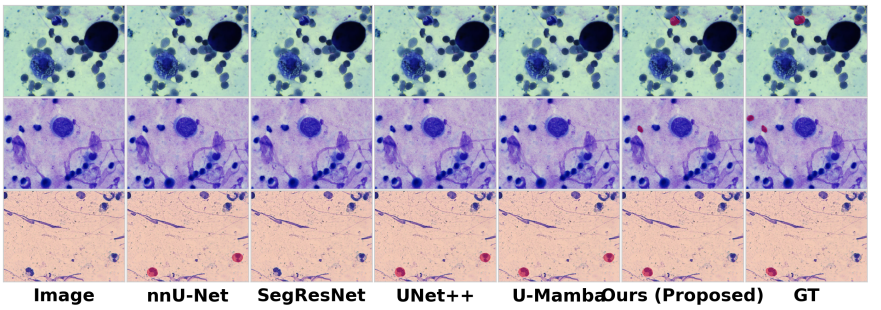

Figure 4: Qualitative comparison of segmentation results. Columns show the input image, predictions from baseline methods (nnU-Net, SegResNet, UNet++, U-Mamba), our proposed method, and the ground truth. Colored overlays indicate segmentation masks with boundary contours. Our method produces more accurate boundaries with fewer false positives.

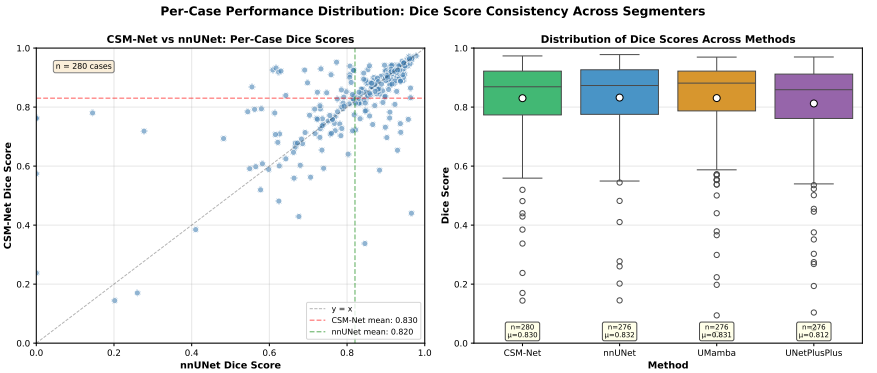

Figure 5: Per-Case Performance Distribution: Dice Score Consistency

Table 7
Computational efficiency comparison on the PW-BALFC dataset.

Method	Params (M)	FLOPs (G)	Inference Time (ms)
nnU-Net	126.56	1019.88	9.6
U-Mamba	104.77	1256.50	19.3
UNet++	2.41	156.11	40.7
SwinU-Mamba	59.89	384.44	44.2
UTNet	10.07	151.03	23.6
UKAN	25.42	60.28	26.3
SwinUNETR	25.16	164.72	28.4
CSM-Net (Ours)	30.75	255.19	20.3

4.3.2. Computational Efficiency

The practical deployment of a segmentation model necessitates a balance between accuracy and computational cost. The efficiency metrics for CSM-Net and baseline models are summarized in Table 7. With 30.75 million parameters and 255.19 GFLOPs, CSM-Net is more lightweight than the leading nnU-Net baseline (126.56M params, 1019.88 GFLOPs). In terms of inference speed, CSM-Net processes an image in 20.3 ms, which is faster than several transformer-based models (e.g., SwinU-Mamba, SwinUNETR) and comparable to other efficient architectures. This favorable trade-off is enabled by the Adaptive Sparse Sampling (ASS) strategy within the MRA blocks, which approximates global cross-scale attention with linear complexity. Consequently, CSM-Net delivers state-of-the-art boundary accuracy while maintaining a practical computational profile suitable for clinical application.

5. Discussion

5.1. Interpretation of Results and Clinical Implications

The experimental evaluation shows that CSM-Net achieves a Dice score of 0.7663, which is statistically equivalent to the top-performing baseline, nnU-Net (0.7666). However, the model's primary advantage lies in its superior boundary delineation, as evidenced by a Hausdorff Distance (95th percentile) of 47.8435, representing an 8.7% improvement over nnU-Net (52.4164). This performance profile is particularly meaningful for the clinical task of BALF cell segmentation. The Dice coefficient measures overall volumetric overlap, which is important for quantifying cell counts and approximate morphology. In contrast, the HD95 metric specifically penalizes the largest segmentation errors at object boundaries. For diagnostic applications, such as identifying atypical cell shapes or quantifying cell membrane irregularities, precise boundary localization is paramount. The significant reduction in HD95 indicates that CSM-Net produces more accurate and consistent cell contours, which can directly translate to more reliable morphological analysis in a clinical workflow. The qualitative results shown in Figure 4 visually confirm this, where our method yields cleaner boundaries with fewer false-positive and false-negative pixels compared to other baselines.

The computational profile of CSM-Net further supports its clinical applicability. With 30.75M parameters and 255.19 GFLOPs, it is substantially more lightweight than the highly parameterized nnU-Net (126.56M parameters, 1019.88 GFLOPs) while maintaining a competitive inference time of 20.3 ms. This efficiency stems from the Adaptive Sparse Sampling (ASS) mechanism, which enables the computationally intensive bidirectional attention of the MRA modules to be performed efficiently on high-resolution feature maps. The balance between high boundary accuracy and practical computational cost makes CSM-Net a suitable candidate for integration into automated diagnostic systems where processing speed and hardware constraints are considerations.

5.2. Analysis of Design Choices and Ablation Studies

The ablation studies provide strong evidence for the effectiveness of the core design principles behind CSM-Net. The removal of the Mutual Refinement Attention (MRA) module led to a drop in Dice score (0.7569 vs. 0.7663) and a degradation in HD95 (48.2420 vs. 47.8435), confirming that our bidirectional cross-scale interaction is a key contributor to performance, especially for boundary precision. The comparison of refinement direction strategies was particularly revealing. The full bidirectional MRA outperformed all unidirectional variants (source-to-target, target-to-source) and alternatives like self-attention only or no attention. This validates our central hypothesis: effective multi-scale representation for this task is not achieved by merely propagating context from coarse to fine scales, but through a mutual, interactive refinement where detailed features inform semantic context and vice versa.

The design of the Adaptive Sparse Sampling (ASS) strategy was also critically validated. Our ASS mechanism outperformed the computationally prohibitive full attention, as well as simpler pooling-based (average, max) or random sampling strategies. This demonstrates that a content-aware, adaptive selection of key tokens based on a combined score of semantic similarity and spatial proximity is more effective than fixed or random approximations. The ablation on the spatial prior weight λ further supports this design, showing that a balanced combination ($\lambda = 0.2$) performs well, while purely content-based ($\lambda = 0.0$) or overly spatially-biased selections lead to performance drops. Finally, the experiment on MRA placement confirmed the intuition behind integrating these modules at intermediate-to-deep network levels (levels 2-3, 3-4, 4-5). This strategic placement allows the model to establish a dialogue across scales where mid-level semantic features and high-resolution details are most effectively negotiated, outperforming placement at only shallow levels, all levels, or solely at the bottleneck.

5.3. Comparison with State-of-the-Art and Limitations

When positioned among contemporary segmentation architectures, CSM-Net carves out a distinct niche. It matches the volumetric segmentation accuracy of the highly engineered and dataset-specific nnU-Net framework, which remains a formidable baseline in medical imaging Isensee et al. (2020). More importantly, it surpasses all listed baselines, including transformer-based (SwinUMamba, UTNet), modern CNN (UNetPlusPlus), and state-space model (UMamba) architectures, in terms of boundary accuracy as

measured by HD95. This suggests that the principle of Cross-Scale Mutual Refinement addresses a specific weakness in existing models when applied to the challenging domain of high-resolution cytology imagery with morphologically similar classes.

However, several limitations of this work must be acknowledged. First, the evaluation was conducted on a single, publicly available dataset (PW-BALFC). While the dataset is substantial and clinically relevant, the generalizability of CSM-Net to BALF samples prepared with different staining protocols, acquired from different microscope models, or containing different cell type distributions remains to be thoroughly validated. Second, the current implementation of ASS, while efficient, introduces additional hyperparameters (e.g., base sparsity ratio β, spatial prior weight λ) that were tuned empirically. A more rigorous, theoretically-grounded method for setting these parameters or making them fully learnable could be explored. Third, the model is designed for 2D image segmentation. Extending the MRA and ASS concepts to 3D volumetric data, such as from confocal microscopy or CT scans, would require careful architectural modifications to handle the increased computational complexity.

5.4. Future Work

The promising results of CSM-Net open several avenues for future research. A direct extension would be to adapt the framework for instance segmentation of BALF cells, which requires not only semantic masks but also separation of individual cell instances. The rich multi-scale features produced by MRA modules could be beneficial for distinguishing touching or overlapping cells. Another important direction is domain generalization and adaptation. The ASS mechanism's ability to focus on salient features could be leveraged to improve robustness across different medical centers and staining variations. Techniques like test-time adaptation or domain-invariant training could be integrated with the MRA blocks.

The core idea of Cross-Scale Mutual Refinement is not limited to cell segmentation. It could be applied to other medical imaging tasks where integrating local detail and global context is crucial, such as gland segmentation in histopathology, lesion delineation in radiology, or instrument segmentation in surgical videos. Exploring the integration of CSM-Net's refinement blocks with other emerging backbone architectures, such as vision Mamba or other efficient long-range dependency models, could lead to further performance gains. The adaptive gating mechanism within MRA could also be made more sophisticated, perhaps conditioned on task-specific metadata or learned to control the flow of information across multiple scales simultaneously.

6. Conclusion

Accurate segmentation of cells in bronchoalveolar lavage fluid microscopy images is a critical yet challenging task for clinical diagnostics, primarily due to the subtle visual distinctions between morphologically similar cell types. To address the limitations of conventional multi-scale segmentation models, which often rely on unidirectional feature propagation and inefficient global context modeling, this work introduced CSM-Net. Our architecture is built upon the principle of Cross-Scale Mutual Refinement, integrating a Mutual Refinement Attention (MRA) mechanism and an Adaptive Sparse Sampling (ASS) strategy within a U-Net-like backbone. This design enables efficient, bidirectional information exchange between feature scales, allowing high-resolution details and low-resolution semantics to mutually refine each other for more discriminative feature learning.

Our comprehensive evaluation on the publicly available PW-BALFC dataset shows that CSM-Net achieves state-of-the-art boundary delineation accuracy, significantly outperforming strong baseline models. While achieving comparable volumetric overlap, the model's key contribution is its superior precision in capturing complex cellular boundaries, which is paramount for accurate morphological analysis. This performance is attained with a computationally efficient design, making the model suitable for potential clinical deployment.

A primary limitation of this study is its validation on a single 2D microscopy dataset. Future work will focus on extending the core cross-scale mutual refinement framework to 3D volumetric segmentation and instance-level segmentation tasks. The generalizability of our MRA and ASS modules should be explored across other medical imaging modalities where precise multi-scale feature integration is essential.

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